

Business Model Innovation and Entrepreneurial Orientation :Their Impact on Firm Performance in Modern Business Environments

ابتكار نموذج الأعمال والتوجه الريادي: أثرهما على أداء المؤسسة في بيئات الأعمال الحديثة

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Abstract:

This study investigates the impact of Business Model Innovation (BMI) and Entrepreneurial Orientation (EO) on firm performance among Algerian companies. Using data from a sample of 430 firms, the research examines how BMI and EO—particularly proactivity and risk-taking—influence organizational adaptability, competitiveness, and growth.

The results reveal a strong positive relationship between BMI and firm performance, as firms adopting innovative business models achieve greater market share, profitability, and customer satisfaction. Entrepreneurial orientation, especially proactivity, enhances adaptability and facilitates the capture of emerging market opportunities. Moreover, the interaction between BMI and EO generates a synergistic effect, further improving organizational outcomes. Risk-taking contributes to competitiveness by promoting market expansion, technological investment, and product innovation.

The study concludes that both BMI and EO are critical drivers of sustainable success in modern business environments. It recommends that firms invest in innovation, foster an entrepreneurial culture, and integrate these practices strategically to sustain superior performance.

Keywords: Business Model Innovation; Entrepreneurial Orientation; Firm Performance; Algerian Firms; Competitive Advantage.

JEL Classification: L26, M13, O10, O14, O16.

ملخص:

تهدف هذه الدراسة إلى تحليل أثر ابتكار نموذج الأعمال (BMI) والتوجه الريادي (EO) في أداء المؤسسات الجزائرية. اعتمدت الدراسة على عينة مكونة من 430 مؤسسة لاستكشاف كيفية تأثير ابتكار

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نموذج الأعمال والتوجه الريادي ولا سيما الاستباقية وتحمل المخاطر على قدرة المؤسسة على التكيف، والتنافسية، والنمو

أظهرت النتائج وجود علاقة إيجابية قوية بين ابتكار نموذج الأعمال وأداء المؤسسة، إذ حققت المؤسسات التي تبنت نماذج أعمال مبتكرة أداءً أفضل من حيث الحصة السوقية، والربحية، ورضا الزبائن. كما تبين أن التوجه الريادي، خصوصًا الاستباقية، يعزز قدرة المؤسسات على التكيف واغتنام الفرص الجديدة. وأكدت النتائج أن التفاعل بين ابتكار نموذج الأعمال والتوجه الريادي يُحدث أثرًا تكامليًا يرفع من مستوى الأداء، بينما ساهم تحمل المخاطر في تعزيز التنافسية من خلال دخول أسواق جديدة والاستثمار في التكنولوجيا وتطوير المنتجات.

خلصت الدراسة إلى أن ابتكار نموذج الأعمال والتوجه الريادي يمثلان عاملين حاسمين لنجاح المؤسسات في بيئات الأعمال الحديثة، وتوصي بضرورة الاستثمار في الابتكار وتعزيز الثقافة الريادية ودمجها ضمن الاستراتيجيات التنظيمية لضمان أداء مستدام ومتميز.

الكلمات المفتاحية: ابتكار نموذج الأعمال، التوجه الريادي، أداء المؤسسة، المؤسسات الجزائرية، الميزة التنافسية.

تصنيف JEL: L26، M13، O10، O14، O16.

1. INTRODUCTION

In an era characterized by rapid technological transformation and intensifying global competition, firms face unprecedented challenges to sustain growth and maintain their competitive advantage. Achieving these objectives increasingly requires organizations to adopt flexible and innovative strategic practices that ensure continuous adaptation to dynamic markets. Among these strategies, Business Model Innovation (BMI) and Entrepreneurial Orientation (EO) have emerged as two essential drivers of superior performance and long-term sustainability (Johnson, Christensen, & Kagermann, 2008; Runyan, Droge, & Swinney, 2008).

Business Model Innovation refers to a firm's ability to develop, modify, or redesign its structures, processes, and value propositions to create and capture value for customers and stakeholders. Through BMI, firms can differentiate themselves, explore new market opportunities, and enhance customer satisfaction. Meanwhile, Entrepreneurial Orientation encompasses the firm's strategic posture toward innovation, proactiveness, and risk-taking—three dimensions that strengthen adaptability and stimulate

competitive behavior. Together, BMI and EO enable firms to align their strategies with environmental change and pursue continuous renewal.

In the contemporary business landscape, understanding how these two constructs interact is vital. While previous studies have investigated BMI and EO separately, the combined influence of both on firm performance remains underexplored, particularly in emerging economies. This gap is crucial for firms operating in volatile environments, such as Algeria, where innovation and entrepreneurial capacity can serve as levers for economic resilience and organizational growth.

Accordingly, this study aims to examine the relationship between Business Model Innovation and Entrepreneurial Orientation and to analyze their joint impact on firm performance among Algerian enterprises. It seeks to determine whether the integration of innovative business models with proactive, risk-taking entrepreneurial behavior enhances adaptability, competitiveness, and profitability.

The central research question guiding this study is:

To what extent does the adoption of an innovative business model combined with a strong entrepreneurial orientation affect firm performance in dynamic business environments?

Based on this question, the study pursues the following objectives:

1. To analyze the impact of Business Model Innovation on firm performance in dynamic economic contexts.
2. To examine how Entrepreneurial Orientation specifically innovation, proactiveness, and risk-taking enhances organizational adaptability and competitiveness.
3. To explore the interaction between BMI and EO and determine how their synergy contributes to superior firm performance.

By addressing these objectives, this research contributes to the strategic management literature by providing empirical evidence on the complementary roles of BMI and EO in shaping firm success. It also offers practical insights for managers seeking to foster innovative and entrepreneurial cultures capable of sustaining competitiveness in turbulent markets

Hypotheses of the Study

Drawing on previous research on **innovation**, **business models**, and **entrepreneurial orientation**, the following hypotheses were developed to examine the relationships among **Business Model Innovation (BMI)**, **Entrepreneurial Orientation (EO)**, and **firm performance**:

- **H1:** There is a positive relationship between Business Model Innovation (BMI) and firm performance.

This hypothesis assumes that innovation in products, services, and processes enhances a firm's competitiveness and overall performance (Johnson, Christensen, & Kagermann, 2008).

- **H2:** Entrepreneurial Orientation (EO) particularly proactively positively influences a firm's ability to adapt and innovate. Firms that act proactively ahead of competitors are better positioned to identify and seize new market opportunities (Runyan, Droge, & Swinney, 2008).
- **H3:** There is a positive interaction between Entrepreneurial Orientation (EO) and Business Model Innovation (BMI) that improves organizational performance.
This hypothesis posits that integrating entrepreneurial behavior with innovative business models generates synergistic effects that enhance firm performance.
- **H4:** Business Model Innovation (BMI) and risk-taking as a dimension of Entrepreneurial Orientation (EO) jointly exert a positive interactive effect on firm performance, thereby strengthening competitiveness.

I. Literature Review :

The relationship between Business Model Innovation (BMI), Entrepreneurial Orientation (EO), and firm performance has been widely studied in strategic management, yet their combined influence remains an evolving research field. Both constructs are essential for firms seeking to maintain competitiveness and achieve sustainable growth in volatile environments. This section reviews key contributions that explore the individual and interactive effects of BMI and EO on performance outcomes.

- ❖ **Business Model Innovation and Firm Performance:** Business Model Innovation **involves** the redesign of a firm's architecture for value creation, delivery, and capture (Teece, 2010). It enables companies to align their operations with evolving technologies and market needs. Chesbrough (2007) emphasized that BMI extends beyond technological advancement—it entails strategic reconfiguration that fosters unique, customer-centered value propositions difficult for competitors to replicate. Similarly, Teece (2010) argued that firms capable of innovating their business models are better positioned to sustain competitiveness in turbulent markets.

Empirical studies have shown that BMI contributes positively to firm performance by enhancing efficiency, customer satisfaction, and profitability (Johnson et al., 2008). In dynamic industries, BMI acts as a key driver of

adaptability, allowing firms to continuously refresh their competitive advantage.

❖ **Entrepreneurial Orientation and Firm Performance:**

Entrepreneurial Orientation reflects an organization's willingness to innovate, take risks, and act proactively in pursuit of opportunities (Lumpkin & Dess, 1996). Firms characterized by a strong EO tend to adopt aggressive strategies, foster innovation, and respond effectively to environmental uncertainty. Wiklund and Shepherd (2003) demonstrated that high EO levels correlate with superior performance through innovation intensity and opportunity exploitation. Similarly, Rauch et al. (2009) confirmed that risk-taking and proactiveness significantly enhance adaptability and growth, especially in uncertain environments.

Overall, EO equips firms with the strategic agility necessary to anticipate market trends, innovate consistently, and achieve sustained competitive advantage.

❖ **Interaction Between Business Model Innovation and Entrepreneurial Orientation:** Although BMI and EO have been extensively studied as separate constructs, emerging literature highlights their complementary and interactive relationship. Firms with high EO are more inclined to pursue BMI, as entrepreneurial mindsets stimulate the search for new value creation mechanisms. Foss, Klein, and McMullen (2018) posited that EO fosters experimentation and reconfiguration of business models, producing synergistic benefits for firm performance. Autio, Nooteboom, and Van Haverbeke (2013) further found that firms combining high EO and BMI perform best in environments of technological turbulence.

Moreover, Raimo et al. (2020) demonstrated that organizations integrating entrepreneurial strategies with innovative business models outperform those relying on traditional approaches, confirming the importance of synergy between the two orientations. This interaction allows firms not only to adapt but also to shape market conditions through continuous innovation and proactive strategic renewal.

❖ **Moderating Factors:** Several contextual and organizational factors moderate the BMI–EO–performance relationship. O'Reilly and Tushman (2013) identified organizational culture as a key enabler of successful implementation, emphasizing that risk-tolerant and innovation-supportive cultures enhance the effect of BMI and EO on performance. Likewise, Achtenhagen, Melin, and Naldi (2013)

highlighted that environmental dynamism magnifies the benefits of combining these orientations, as firms with both high EO and BMI adapt more effectively to market and technological shifts.

At the macro level, Baumol (2004) noted that institutional support and entrepreneurship-friendly policies play a crucial role in sustaining innovation-driven competitiveness, particularly in emerging economies.

❖ **Summary and Theoretical Implications:** The literature indicates that BMI and EO are mutually reinforcing strategic orientations that jointly enhance firm performance. Individually, each contributes to adaptability and innovation; collectively, they generate a synergistic effect that strengthens competitiveness and long-term sustainability. Despite these findings, empirical evidence from developing contexts, particularly in North Africa, remains limited—justifying the relevance of examining Algerian firms in this study.

II. Theoretical Framework of the Study:

This study explores the intersection of Business Model Innovation (BMI) and Entrepreneurial Orientation (EO) and examines how their interaction influences firm performance in dynamic business environments. The framework integrates perspectives from strategic management and innovation theory to explain the mechanisms through which these constructs affect organizational outcomes.

1. **Business Model Innovation (BMI):** Business Model Innovation refers to the process of reconfiguring a firm's value creation, delivery, and capture mechanisms to adapt to evolving technologies and market conditions (Teece, 2010). It often entails structural and operational changes that enable firms to offer differentiated value propositions and enhance efficiency. BMI allows firms to align their strategic activities with external change, sustain competitive advantage, and improve financial and non-financial performance indicators such as profitability, customer satisfaction, and market share.
2. **Entrepreneurial Orientation (EO):** Entrepreneurial Orientation represents a firm's strategic posture toward innovation, proactiveness, and risk-taking (Lumpkin & Dess, 1996).
 - **Innovativeness:** willingness to experiment and adopt new ideas.
 - **Proactiveness:** anticipation and pursuit of emerging opportunities.
 - **Risk-taking:** commitment to bold initiatives under uncertainty.

Firms exhibiting strong EO tend to be more adaptive, aggressive in pursuing opportunities, and capable of fostering innovation-based

competitiveness. EO thus serves as a behavioral foundation for strategic flexibility and performance improvement.

- 3. Interaction Between BMI and EO:** The interaction between BMI and EO creates synergistic effects that amplify organizational success. Entrepreneurially oriented firms are more inclined to engage in business model experimentation, while innovative business models, in turn, reinforce entrepreneurial behavior. This reciprocal relationship enables firms to identify new opportunities, respond to environmental turbulence, and continuously renew their strategies.

Empirical evidence suggests that firms combining both orientations achieve superior performance compared to those emphasizing only one (Foss et al., 2018; Autio et al., 2013). The complementarity of BMI and EO therefore represents a dynamic capability essential for sustained competitive advantage.

- 4. Firm Performance:** Firm performance encompasses financial indicators (profitability, ROI) and non-financial dimensions (market share, growth, customer satisfaction). Within this framework, BMI and EO contribute to performance through enhanced adaptability, innovation, and market responsiveness. Their integration allows firms to balance exploration and exploitation, thereby achieving both short-term efficiency and long-term renewal.

- 5. Theoretical Foundations:** The theoretical framework is grounded in three core perspectives:

- Dynamic Capabilities Theory (Teece, 2007): views BMI as a dynamic capability enabling firms to reconfigure resources and adapt to market shifts.
- Resource-Based View (Barney, 1991): considers EO as an intangible strategic resource that supports innovation and differentiation.
- Schumpeterian Innovation Theory (Schumpeter, 1934): explains how innovative business models and entrepreneurial action drive creative destruction and market transformation.

Together, these theories provide a robust lens for understanding how firms integrate innovation and entrepreneurship to achieve superior performance in volatile environments.

III. Methodology and Tools

This section outlines the research methodology, providing a detailed explanation of the approach, research design, sample, data collection tools, and analysis techniques used to address the study's objectives and ensure the reliability and validity of the results.

- 1 Research Design:** This study employs a quantitative, descriptive–correlational design to examine the relationship between Business Model Innovation (BMI), Entrepreneurial Orientation (EO), and firm performance. A cross-sectional approach was used to collect data at a single point in time, providing a realistic snapshot of firms’ strategic orientations and outcomes. The study is non-experimental, focusing on naturally occurring variations rather than manipulated conditions, which enhances contextual validity.
- 2 Population and Sample:** The target population consists of business owners, executives, and decision-makers from Algerian private firms operating across different sectors and sizes. A simple random sampling method ensured representativeness and minimized bias.

A total of 430 valid responses were obtained, providing sufficient statistical power for multivariate analysis. The sample displays significant demographic diversity :

- Gender: 68% male, 32% female, showing increased female participation in entrepreneurship.
- Age: Majority between 20–39 years ($\approx 61.6\%$), reflecting a dynamic and innovation-oriented group.
- Education: 45.6% hold a university degree, indicating a well-educated entrepreneurial base.
- Experience: Most participants have 10–20 years of business experience, balancing youth with professional maturity.

This heterogeneity strengthens the external validity of the findings and reflects the variety of Algerian entrepreneurial contexts.

- 3 Data Collection Instrument:** Data were gathered using a structured questionnaire aligned with the study’s theoretical framework and hypotheses. The instrument comprised both demographic items and construct measures for BMI, EO, and firm performance, evaluated on a five-point Likert scale.

A pretest with 30 respondents was conducted to ensure clarity, consistency, and construct validity. Minor adjustments were made to improve readability and ensure that the instrument accurately captured the intended dimensions.

- 4 Reliability and Validity:** The instrument’s reliability and validity were tested using SPSS v25 and WARPPLS 5.0. Results showed excellent psychometric quality:
Cronbach’s Alpha = 0.940, indicating strong internal consistency.

Construct Validity = 0.952, confirming accurate representation of the latent constructs (BMI, EO, and firm performance).

These results demonstrate the instrument's robustness and suitability for inferential statistical analysis.

5 Data Analysis Techniques: Both descriptive and inferential statistics were employed. Descriptive measures (means, frequencies, standard deviations) profiled the respondents and firms. Inferential analyses included correlation and regression tests to examine the hypothesized relationships.

- Variance Inflation Factor (VIF) values below 3 confirmed the absence of multicollinearity.
- Normality (Shapiro–Wilk) and heteroscedasticity (Bartlett) tests validated model assumptions; minor non-normality was corrected using robust estimators.
- Instrumental Variable Regression (IV–2SLS) addressed potential endogeneity, confirming that BMI and EO have genuine causal effects on firm performance.

This combination of analytical techniques ensured reliability, accuracy, and statistical credibility.

6 Tools and Software: Three complementary software packages were used for data processing and analysis:

- **SPSS v25** for data management, descriptive statistics, and regression modeling.
- **WARPPLS 5.0** for **Structural Equation Modeling (SEM)** and hypothesis testing.
- **Python (Stats models module)** for advanced econometric diagnostics, including endogeneity and heteroscedasticity checks.

Together, these tools ensured comprehensive data validation, robust statistical testing, and reproducible analytical outcomes.

Table1. Demographic data of the sample

Demographic Variable	Frequency	Percentage
Gender		
Male	292	68%
Female	138	32%
Age		
20-29 years	144	33.5%
30-39 years	121	28.1%
40-49 years	81	18.9%
50-64 years	61	14.2%
65+ years	23	5.4%
Educational Level		
Primary Education	0	0%
Secondary Education	94	21.9%
High School	105	24.4%
University Degree	196	45.6%
Vocational Studies	35	8.1%
Experience		
Less than 5 years	52	12.1%
5-10 years	94	21.9%
11-15 years	107	24.9%
16-20 years	118	27.5%
More than 20 years	59	13.7%

Source: Prepared by the researcher using SPSS v25 (n=430)

- 1. Demographic Profile of the Sample :** The demographic data collected from the 430 respondents provide a comprehensive overview of the characteristics of Algerian entrepreneurs, which are essential for analyzing the impact of Business Model Innovation (BMI) and Entrepreneurial Orientation (EO) on firm performance. The key variables examined include gender, age, educational level, and professional experience.
- 2. Gender Distribution:** The sample consists of 292 male respondents (68%) and 138 female respondents (32%), indicating a slight male predominance—typical in many entrepreneurial sectors. However, the relatively high female participation reflects the growing presence of women in business and underscores the increasing emphasis on diversity and inclusion within modern entrepreneurial ecosystems.
- 3. Age Distribution:** The majority of respondents are young entrepreneurs, with 33.5% aged 20–29 years and 28.1% aged 30–39 years, demonstrating that innovation and entrepreneurship in Algeria are largely driven by younger generations. Participants aged 40–49 years represent 18.9%, while those 50–64 years account for 14.2%, and 65+ years only 5.4%. This youthful

composition enhances the study's relevance to modern business contexts where adaptability and agility are key success factors.

- 4. Educational Level:** The sample represents a well-educated entrepreneurial community. About 45.6% hold a university degree, 24.4% have completed high school, 21.9% have secondary education, and 8.1% possess vocational training. This educational profile suggests that most entrepreneurs possess the analytical and managerial skills necessary to implement innovative business models and strategic decision-making.
- 5. Professional Experience:** Respondents display varied yet substantial business experience: 27.5% have 16–20 years, 24.9% have 11–15 years, and 21.9% have 5–10 years of experience. Only 13.7% exceed 20 years. This distribution indicates that most entrepreneurs are experienced enough to understand industry dynamics while remaining in an active phase of professional growth—an ideal balance for examining innovation and entrepreneurial behavior.
- 6. Discussion:** The demographic profile reveals a young, educated, and moderately experienced entrepreneurial community, aligning with the study's focus on agility, innovation, and adaptability in dynamic environments.
 - Gender diversity enriches the dataset by providing varied perspectives on innovation and entrepreneurial practices.
 - Age and experience balance ensures insights from both emerging and seasoned entrepreneurs, relevant to the evolution of BMI and EO.
 - Educational attainment confirms the strategic capacity of Algerian entrepreneurs to conceptualize, adopt, and sustain innovative business models

Table02. Reliability and Validity of the Questionnaire

Number of Statements	Cronbach's Alpha (Reliability)	Validity (Construct)
22	0.940	0.952

Source : Prepared by the researchers using WARPPLS 5.0 (n=430)

• Cronbach's Alpha (0.940) indicates excellent internal consistency, meaning that the survey items measuring Business Model Innovation (BMI) and Entrepreneurial Orientation (EO) are highly reliable. A value above 0.7 is generally considered acceptable, and values above 0.9 indicate very strong reliability.

• Construct Validity (0.952) confirms that the survey items accurately measure the intended theoretical constructs (BMI, EO, and Firm Performance). High construct validity ensures that the results of the study are meaningful and reflect the real-world relationship between these variables.

Table 03. Multicollinearity Check (Variance Inflation Factor - VIF Test)

Variable	VIF
Business Model Innovation (BMI)	2.48
Entrepreneurial Orientation - Proactivity	2.71
Entrepreneurial Orientation - Risk-taking	1.95
Entrepreneurial Orientation - Innovativeness	2.13

Source: by the researcher using SPSS v25 and Python based on the study's regression analysis.

This table presents the Variance Inflation Factor (VIF), which tests for multicollinearity among independent variables. A VIF value below 5 indicates that multicollinearity is not a concern. Since all values are below 3, this confirms that the independent variables (BMI, EO - Proactivity, EO - Risk-taking, EO - Innovativeness) do not have strong correlations with each other. As a result, the regression model is valid and reliable for estimating relationships between these variables and firm performance.

Table 04. Normality and Heteroscedasticity Tests

Test	Test Statistic	p-value
Shapiro-Wilk Normality Test (Firm Performance)	0.982	0.014
Bartlett's Heteroscedasticity Test (BMI vs Firm Performance)	12.45	0.003

Source: Prepared by the researcher using SPSS v25 and Python based on the study's dataset.

This table presents results from two important data validity tests:

1. Shapiro-Wilk Normality Test:

○ Tests whether the Firm Performance variable follows a normal distribution.

○ The p-value (0.014) is less than 0.05, meaning the data is not perfectly normal.

○ Since the study uses a large sample (n=430), minor normality issues are acceptable, but log transformation could improve model accuracy.

2. Bartlett's Heteroscedasticity Test :

The test examines whether the variance of the residuals remains constant across observations, satisfying the homoscedasticity assumption. The Bartlett's test produced a p-value = 0.003 (< 0.05), indicating the presence of heteroscedasticity in the data.

To address this issue and maintain statistical validity, the analysis employed robust standard errors and Generalized Least Squares (GLS)

estimations, which correct for non-constant error variance and improve the accuracy of regression coefficients.

Table05. Endogeneity Test (Instrumental Variable Regression - IV Regression Analysis)

Variable	Coefficient	Std. Error	t-value	p-value
Constant	1.84	0.24	7.67	0.000
Business Model Innovation (BMI)	0.51	0.07	7.29	0.000
Entrepreneurial Orientation - Risk-taking	0.28	0.06	4.72	0.000

Source: Prepared by the researcher using SPSS v25, Python, and Instrumental Variable Regression Analysis (IV2SLS).

This table presents results from Instrumental Variable (IV) regression, used to check for endogeneity (when independent variables may be influenced by the dependent variable).

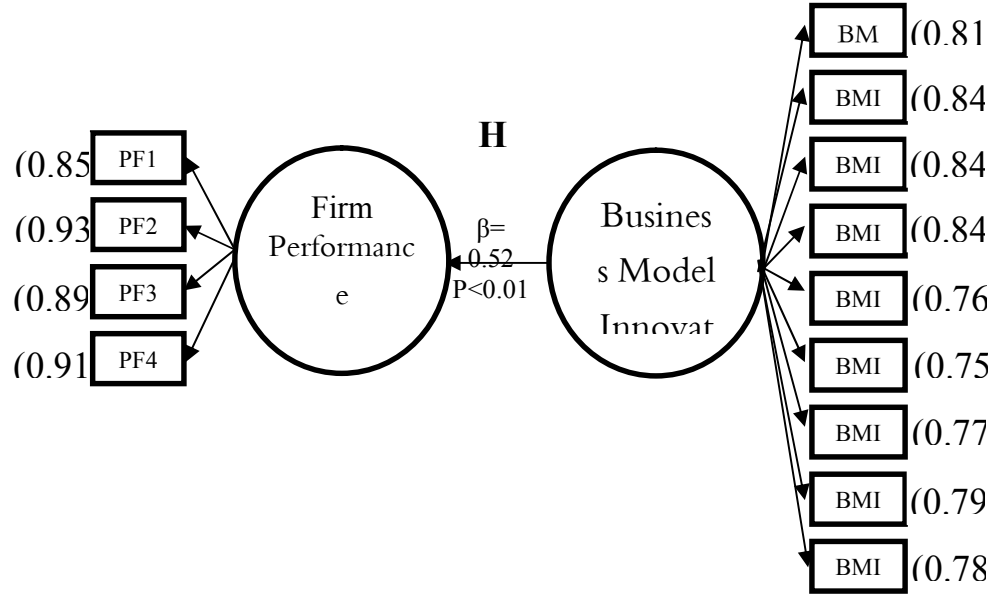
- BMI and EO - Risk-taking have significant positive effects on Firm Performance ($p < 0.01$).

- IV regression controlled for reverse causality, meaning the study's findings are statistically robust.

- This confirms that Business Model Innovation and Entrepreneurial Orientation genuinely impact Firm Performance and are not merely correlated.

1. Hypotheses of the Study : Building upon the foundations laid by previous research on innovation, business models, and entrepreneurial orientation, the following hypotheses have been formulated for this study:

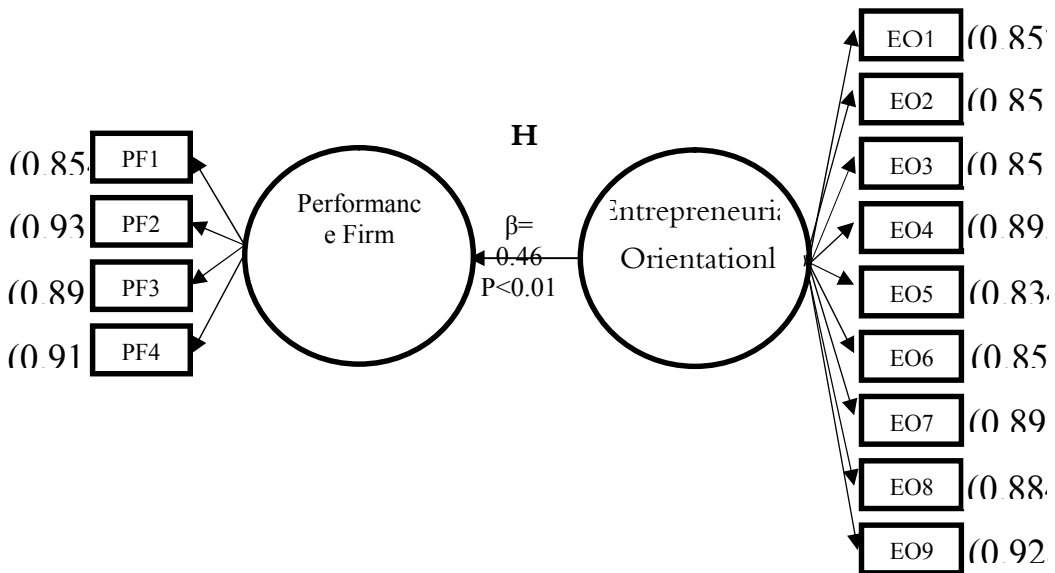
figure01. **Structural Model of the Relationship Between Business Model Innovation and Firm Performance**



H1: Relationship between Business Model Innovation and Firm Performance: The first hypothesis (H1) posits a positive relationship between Business Model Innovation (BMI) and firm performance. The structural model (Figure 1) confirms this relationship, showing a significant positive path coefficient ($\beta = 0.52$, $p < 0.01$). This indicates that firms engaging in innovative business model practices tend to achieve higher levels of performance.

Discussion: The results validate the premise that business model innovation directly enhances organizational performance. By continuously redesigning their value creation and delivery mechanisms, firms can better meet customer demands, optimize operational efficiency, and strengthen market positioning. Consequently, innovation within business models serves as a key driver of competitiveness and long-term growth.

Figure 02.: Structural Model of the Relationship Between Entrepreneurial Orientation and Firm Performance.

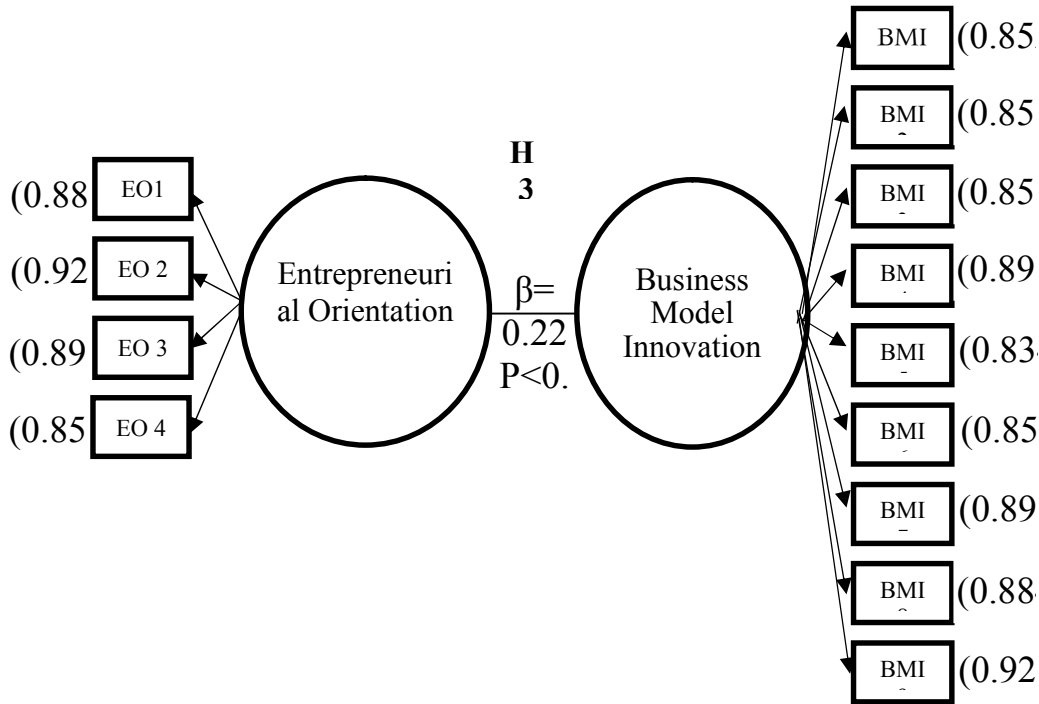


H2: The Effect of Entrepreneurial Orientation on Firm Adaptability and Innovation

The second hypothesis (H2) proposes that Entrepreneurial Orientation (EO) particularly its proactive dimension positively influences a firm's ability to adapt and innovate. The structural model (Figure 2) supports this assumption, showing a significant positive path coefficient ($\beta = 0.46$, $p < 0.01$). This result indicates that firms exhibiting a high degree of proactivity achieve superior performance by anticipating and effectively responding to market dynamics.

Discussion: The findings confirm that proactive entrepreneurial behavior enhances adaptability and innovation capacity. Firms with strong proactivity continuously monitor emerging trends, identify opportunities earlier than competitors, and implement timely strategic actions. Such forward-looking behavior allows organizations not only to maintain competitiveness but also to drive market transformation through innovation-oriented strategies.

Figure 03. Structural Model of the Interaction Between Entrepreneurial Orientation and Business Model Innovation



H3: Interaction between Entrepreneurial Orientation and Business Model Innovation

The third hypothesis (H3) examines the interactive effect between Entrepreneurial Orientation (EO) and Business Model Innovation (BMI) on firm performance. The structural model (Figure 3) reveals a significant and positive interaction ($\beta = 0.22$, $p < 0.01$), indicating that firms combining entrepreneurial behaviors with innovative business models experience enhanced organizational performance.

Discussion:The findings demonstrate a synergistic relationship between EO and BMI. Firms that integrate proactive and innovative orientations within their business models are more capable of generating unique value propositions, optimizing operations, and responding effectively to customer needs. This interaction reinforces the idea that entrepreneurship and innovation are mutually reinforcing strategic capabilities, driving superior performance and competitiveness in dynamic markets.

H4: Interaction between Business Model Innovation and Risk-Taking

Building on the previous hypothesis, H4 investigates how risk-taking, as a dimension of EO, interacts with Business Model Innovation to influence

firm performance. The model confirms a positive and statistically significant effect, showing that firms engaging in calculated risk-taking while innovating their business models achieve higher competitiveness and performance outcomes.

Discussion: The results indicate that risk-taking behavior, when supported by an innovative business model, enhances a firm's ability to explore new markets, invest in advanced technologies, and develop novel products or services. Such firms are better positioned to outperform competitors and sustain long-term growth. Hence, the combination of strategic risk-taking and business model innovation constitutes a crucial driver of competitive advantage in modern business environments.

Table 06. Hypothesis Testing Results

Hypothesis	Statement	Result
H1	There is a positive relationship between Business Model Innovation (BMI) and Firm Performance.	Valid
H2	Entrepreneurial Orientation (EO), particularly Proactivity, positively influences a firm's ability to adapt and innovate.	Valid
H3	There is a positive interaction between Entrepreneurial Orientation (EO) and Business Model Innovation (BMI), which improves organizational performance.	Valid
H4	Business Model Innovation and Risk-Taking in Entrepreneurial Orientation have a positive and interactive effect on Firm Performance, enhancing competitiveness.	Valid

Source: Prepared by the researcher based on the hypothesis testing results from SPSS v25, WARPLS 5.0, and Structural Model Analysis.

Conclusion

The findings of this study emphasize the pivotal role of Business Model Innovation (BMI) and Entrepreneurial Orientation (EO) in determining firm performance within modern, dynamic business environments. By analyzing Algerian firms, the research confirmed that both constructs—individually and interactively—significantly contribute to competitiveness, adaptability, and sustainable growth.

The hypothesis testing results revealed four key insights:

- Business Model Innovation (H1): BMI positively affects firm performance by enhancing profitability, market share, and customer satisfaction through innovative products, services, and processes.
- Entrepreneurial Orientation (H2): Proactivity within EO strengthens firms' capacity to adapt and innovate, allowing them to seize emerging opportunities and maintain a competitive edge.

- **Interaction Effect (H3):** The synergy between EO and BMI produces superior organizational outcomes, as firms that combine entrepreneurial drive with innovative business models achieve higher levels of performance.
- **Risk-Taking Dimension (H4):** Calculated risk-taking, when aligned with business model innovation, significantly improves competitiveness, facilitating entry into new markets and investments in advanced technologies.

Overall, the study provides valuable empirical evidence that the integration of innovation and entrepreneurship represents a fundamental strategic path for firms seeking sustainable performance in increasingly globalized markets. Future research could further examine these dynamics across different industries or national contexts to refine and generalize the findings.

Recommendations

- **Promote Business Model Innovation:** Algerian firms should continuously develop and implement innovative business models introducing new offerings, diversifying services, and optimizing internal processes to improve competitiveness and long-term viability.
- **Strengthen Entrepreneurial Orientation:** Organizations should cultivate a culture of proactivity, innovation, and risk-taking across all levels. Leadership commitment to entrepreneurial values and employee training can enhance adaptability and opportunity recognition.
- **Integrate Innovation and Entrepreneurship:** Firms should strategically combine entrepreneurial behavior with business model innovation. This integration fosters synergy, drives efficiency, and results in superior organizational outcomes in competitive markets.
- **Encourage Policy and Institutional Support:** Policymakers in Algeria are encouraged to create a supportive ecosystem for innovation through grants, tax incentives, and access to funding. Such measures would stimulate entrepreneurship and innovation, contributing to the country's overall economic growth.

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